

SPECops: Spectral Explainability and Circuit Optimization for Quantum-Assisted Physics-Informed Neural Networks

Qinetic Research Lab | WISER $\times$ BQP Industry Track Challenge

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Abstract

Our report aims to answer a question most quantum assisted physics informed neural network (QAP-INN) benchmarks look over: when a quantum layer improves accuracy, does it result from genuine expressivity gain, or is it from another part of the setup? Using the viscous Burgers' equation as our primary benchmark, we derive each configuration's accessible Fourier frequency range from first principles and see if it predicts, quantitatively, where a QAPINN's accuracy degrades. We trace the mechanism to a measurable redundancy signature in the trained circuit, confirmed by two independent metrics via a clean pre-layer control, shown to originate in the quantum layer rather than the input. We do further testing: testing circuit structure variants beyond reupload counts shows how redundancy decreased by removing entanglement correlates with worse accuracy, the opposite direction from redundancy driven down by increasing the reupload count, indicating representational redundancy is not a single, uniform phenomenon with one accuracy consequence. Furthermore, we extended the mechanism to a second PDE, being the heat equation, and find that a large portion of what looked like a quantum accuracy advantage was actually a training budget artifact: under the one optimizer and one epoch matched comparison we ran, classical outperformed quantum. We report this as a fact, and treat the self correction as a part of the finding, offering both the mechanism and the correction as a methodological caution for fixed budget QML benchmarking generally. **Repository Link:** <https://github.com/JitheshMithra/SPECops/>.

1 Introduction

Quantum-Assisted Physics-Informed Neural Networks (QAPINNs) replace the first hidden layer of a classical Physics-Informed Neural Network (PINN) with a variational quantum circuit (VQC). BQP's own benchmarking work [1] showed that this substitution can reduce trainable parameter counts while maintaining accuracy on computational fluid dynamics (CFD) problems. What this work does not explain is *why* this substitution works when it works, or what specifically is gained and lost by replacing a classical layer with a quantum one.

This is the question the challenge poses, and the one this report addresses: not whether QAPINNs outperform PINNs, but when, why, and how the introduction of a quantum layer changes the learning dynamics of a physics-informed network.

We study this on the one-dimensional viscous Burgers' equation,

$$u_t + u u_x = \nu u_{xx}, \quad \nu = 0.01/\pi \tag{1}$$

on the domain $x \in [-1, 1]$, $t \in [0, 1]$, with initial condition $u(x, 0) = -\sin(\pi x)$ and Dirichlet boundary conditions $u(\pm 1, t) = 0$. This is the same benchmark problem used in BQP's original QAPINN paper and in Raissi, Perdikaris & Karniadakis's foundational PINN work [2], chosen specifically because its solution develops a sharp shock as $t \rightarrow 1$, giving a controlled setting in which to probe expressivity limits

Our approach applies a disentanglement methodology recently proposed for a different problem class (fitting the Lorenz-63 attractor, a chaotic attractor meant to simulate the flow of fluid or air heated from below and cooled from above, with a fixed encoding VQC) to this CFD setting: separating whether a QAPINN’s performance is limited by the *expressivity* of its data-encoding scheme (a hard architectural ceiling) or by *trainability* failure (barren-plateau-style vanishing gradients). We extend this methodology with a representational-redundancy analysis that gives a mechanistic account of *how* the expressivity ceiling manifests inside the network, not just that it exists.

We report our results, including a finding that revised our original hypothesis, and a still-open question about the classical vs quantum comparison that we address transparently rather than resolving prematurely.

2 Background and Related Work

2.1 Physics-Informed Neural Networks

Raissi, Perdikaris & Karniadakis (2017) [2] introduced PINNs, in which a feedforward network’s loss function is constructed directly from a PDE’s residual, the residual was computed via automatic differentiation of the network’s own output with respect to its inputs, plus boundary and initial condition terms.

2.2 QAPINN architecture

Shah, Lineswala & Chopra (QCE 2024) [1] introduced the Quantum-Assisted PINN, replacing a PINN’s first hidden layer with a VQC, and benchmarked it on Burgers’ equation across 3, 4, and 5-qubit variants, reporting maintained or improved accuracy with up to 20% fewer trainable parameters. This is the architecture and benchmark we build directly on.

2.3 Expressivity of variational quantum circuits

Schuld, Sweke & Meyer (2021) [3] showed that a variational quantum circuit’s output can be written as a truncated Fourier series in its classical input, with the *accessible frequency spectrum* determined by the data encoding gates. It also showed Data re-uploading, which in this case was repeating the encoding block, extends the accessible frequency range. This is the theoretical anchor of our expressivity analysis.

2.4 Barren Plateaus

McClean, Boixo, Babbush, Smelyanskiy & Neven (2018) [4] showed that for a wide class of parameterized quantum circuits, gradient magnitude vanishes exponentially in qubit count, making the training intractable at scale. We use the gradient variance as our diagnostic for this effect.

2.5 Capacity-ceiling vs trainability disentanglement

A recent analysis of fixed-encoding VQCs applied to the Lorenz-63 attractor that demonstrates apparent training failure can result from a hard architectural capacity ceiling rather than a barren plateau, it also provides a methodology for distinguishing between the two [5]. We apply this same disentanglement to a CFD PINN setting.

2.6 Adjacent quantum-PINN work

QCPINN (2025) systematically evaluates the topology and embedding choices for quantum-classical PINNs, reports comparable accuracy to classical baselines at 10-30% of the parameter count. In our work, rather than taking a broad architecture search, we hold our architecture fixed to isolate *why* a specific design parameter (i.e. re-upload count) affects performance [6], down to a representational mechanism inside our trained network.

2.7 Spectral bias in PINNs.

A separate line of work shows that PINNs can minimize PDE residuals in an averaged sense while failing to make a fine-scale solution structure, in other words shifting towards smooth or trivial solutions under strong nonlinearity [7]. Section 6 shows that this same failure occurs in QAPINNS and ties this phenomenon directly to the quantum layer’s expressivity ceiling.

3 Contribution Statement

This work is an **application and extension** of recently published methodology, specifically the capacity-ceiling/barren-plateau disentanglement framework and the effective-rank expressivity metric, to a new problem domain, CFD-relevant and shock-forming PDEs, rather than a completely new theoretical framework. Our contribution is in the disentanglement applied to this setting, the representational-redundancy mechanism we identify as the substrate of the expressivity ceiling, and the finding that a fixed training budget comparison overstates the role of expressivity relative to optimization dynamics.

4 Method

4.1 Architecture

The QAPINN consists of a classical linear + tanh pre-layer (mapping physical inputs (x, t) to encoding angles $\theta \in [-\pi, \pi]$), a VQC (RY angle encoding, which we repeat for data re-uploading; RY/RZ trainable rotations; CNOT entangling ring), a full-qubit Pauli-Z measurement (every qubit measured, not a single qubit, seen in section 4.2), and a classical post-layer producing the scalar output $u(x, t)$.

4.2 A note on measurement

Our initial implementation measured on a single qubit, we identified and corrected this: doing this bottlenecks the entire quantum layer’s information content to a single scalar regardless of the width of the circuit. It is important to note that all reported results use the full-qubit readout.

4.3 Sweep design

We train $n_{\text{qubits}} \in \{3, 4, 5\}$ crossed with $n_{\text{reuploads}} \in \{1, 2, 3, 5\}$ (12 configurations), 3 seeds each using PennyLane and a PyTorch backend.

4.4 Classical control

A pure classical PINN without a quantum layer, but identical in every other parameter (data, loss, optimizer). We also constructed a parameter-matched classical control to address a capacity asymmetry (Section 9).

4.5 Evaluation

We use the Cole-Hopf analytical solution to Burgers’ equation as ground truth, computed through numerical integration, and report relative L_2 error and PDE residual error on a held-out evaluation grid, separately from the training loss.

4.6 Reproducibility

Two implementation bugs were identified and fixed while testing: (1) the optimizer step was silently a no-op in early runs, which invalidated all pre-fix results (archived separately, not used here); (2) the pre-layer to encoding angle derivative used for physical-frequency conversion (Section 5.2) was initially computed from trained, per-run checkpoints rather than a fixed initialization, confounding a structural quantity with training noise. Both are corrected in all results reported here. All code, data, and reproduction instructions are included in the accompanying repository.

5 Expressivity: The Fourier Ceiling

5.1 Theta-space result

Across all 12 configurations, the maximum accessible Fourier frequency of the quantum layer, measured in units of the encoding angle θ , is exactly $n_{\text{reuploads}}$ (1,2,3, or 5) and is **independent of qubit count**. This matches Schuld et al.’s theoretical prediction exactly, being that each re-upload contributes one accessible frequency, and qubit count alone does not expand the frequency spectrum in the encoding scheme.

5.1.1 Derivation, specific to this circuit

This is not just a match to Schuld et al.’s general theorem; it follows directly from the gate structure used here. Each re-upload layer applies one $\text{RY}(\theta)$ encoding gate per wire. $\text{RY}(\theta) = \exp(-i\theta Y/2)$ has generator $Y/2$ with eigenvalues $\{-1/2, +1/2\}$, and a data-encoding gate’s eigenvalue spectrum determines the frequencies it contributes to the circuit’s output [3]: one such gate contributes exactly one frequency component, $\omega = \pm 1$. Stacking r re-upload layers sums these into the accessible frequency set $\{-r, \dots, -1, 0, 1, \dots, r\}$, giving maximum frequency $= r$ exactly. Crucially, this derivation never references n_{qubits} : adding wires adds parallel copies of the same single-frequency gate, not new frequency components, since each wire’s RY spectrum is generated independently of the others, this is why the theta-space ceiling in Table 5.1 is identical across $q3/q4/q5$ for fixed r , a direct consequence of the gate algebra rather than a numerical coincidence. Qubit count instead expands the *coefficient space* over that fixed frequency basis, the direct mechanistic link to Section 6: more qubits without more reuploads means more wires reporting near-duplicate projections onto the same small basis, not new frequency content.

n_{qubits}	$r=1$	$r=2$	$r=3$	$r=5$
3	1	2	3	5
4	1	2	3	5
5	1	2	3	5

Table 1: Maximum accessible Fourier frequency (theta-space), by qubit count and re-upload count. Identical across qubit count for fixed r .

5.2 Physical- x conversion

Because the circuit operates on the encoding angle θ , not the physical coordinate x , we convert using the pre-layer’s local derivative $d\theta/dx$, computed from a fixed (untrained) initialization to avoid confounding the structural quantity with training-run noise (see Section 4). Since the θ -space ceiling is identical across qubit counts, the small residual variation across qubit counts in the converted physical- x ceiling ($q3$: 1.259 at $r=5$; $q4$: 1.055; $q5$: 1.233) is attributable to per-wire initialization noise in $d\theta/dx$, not a genuine qubit-count effect, confirmed by identical observed minimum/maximum $d\theta/dx$ values across nested wire sets.

5.2.1 Caveat: per-wire spread

The per-wire spread in $d\theta/dx$ is substantial and worth stating plainly: within a single configuration, individual-wire values range from a minimum of 0.088-0.15 to a maximum of ≈ 0.438 , a $3\text{-}5\times$ spread across wires of the *same* circuit. The near constant scaling check (constancy of $d\theta/dx$ across the physical- x domain, for a fixed wire) passes for every configuration, but constancy *across wires* does not hold. The single physical- x ceiling reported per configuration (Section 5.4 and throughout) is therefore a mean over wires with meaningfully different individual ceilings, not an exact quantity we report the mean because it is the only tractable summary for the sufficiency comparison that follows, but flag it explicitly as an approximation.

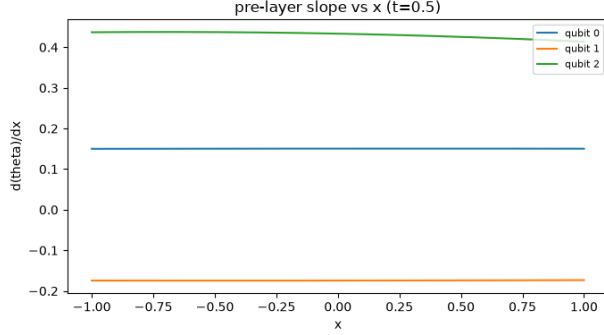


Figure 1: $d\theta/dx$ vs. physical x at $t=0.5$, one line represents one wire. Each wire has a flat slope across x confirming the near constant scaling check within a wire. Each wire differs by roughly $3\text{-}5\times$, the spread discussed above.

5.3 Required bandwidth of the target solution

We computed the Fourier spectrum of the Cole-Hopf reference solution at five time snapshots. The 99%-energy bandwidth grows sharply: approximately 1 cycle at $t = 0.1$, 3 at $t = 0.3$, then 11 at $t = 0.5$ and remaining at 11-11.5 through $t = 0.9$, this shows how most of the required bandwidth growth occurs during shock formation between $t = 0.3$ and $t = 0.5$.

5.4 Sufficiency analysis

Comparing the required bandwidth to the expressivity ceiling per region (smooth: $t < 0.3$; shock: $t \geq 0.5$) across all 12 configurations, under the corrected physical- x units: **no tested configuration reaches expressivity sufficiency in either region**. The best tested configuration ($n_{\text{reuploads}} = 5$) sits at a smooth-region ratio of 1.6-1.9 (required bandwidth over ceiling) and a shock-region ratio of 9-11 both above 1, showing how both regions remain under resourced. The shock region is consistently about $5\times$ further from sufficiency than the smooth region across every configuration tested.

No configuration crosses into sufficiency in either region. The correct claim is a **relative-distance** one: the shock region is proportionally further from expressivity sufficiency than the smooth region, and this relative gap is what the subsequent accuracy analysis in Section 8 must be checked against.

5.5 Structural generalization check: does the derivation extend beyond Burgers' PDE

Section 5.1.1's derivation mainly depends on the encoding circuit's gate structure, not which PDE is being solved, because of this, we checked directly whether it extends to the heat equation ($(u_t = \alpha u_{xx})$, this was another PDE mentioned in the challenge brief which could be used as a generalization test. In our test, we do not reimplement a parallel circuit that could have diverged in the Burgers' test, instead it directly reuses the same `build_model()` output, the same quantum circuit qnode object, the same pre-layer and its scaling, only the PDE residual and loss function differ between the two scripts. Because the encoding circuit is the same object in both cases, Section 5.1.1's derivation applies to the heat equation encoding with no adjustment and no residual uncertainty about implementation drift.

We subsequently completed the empirical half of this check, by running a full 12 configuration sweep on the heat equation, at two seeds each. The single seed pass alone was noisy and non monotonic in both qubit count and reupload count (i.e., $q5, r=2$ at $L_2=0.794$, worse than $q5, r=1$'s 0.610), initially looking consistent with our theory. However, a second seed resolved this for some configurations, $q5, r=2$ drops to 0.249, consistent with the general pattern elsewhere in the sweep, confirming the first seed's value there was noise. It does not resolve every configuration: $q5, r=1$ moves from 0.610 to 0.674 and $q5, r=5$ from 0.258 to 0.346, being seed to seed movement rather than convergence. Two seeds are enough to show the single seed sweep was unreliable, not enough to call the cross PDE empirical validation settled.

The same correlation and effective rank diagnostics from Section 6.1/6.1.1, run against the heat equation checkpoints (in both seeds), reproduce the redundancy signature: $r=1$ configurations cluster at near total redundancy (0.92-1.00 across both seeds, all qubit counts), while $r \geq 2$ configurations show more variation and usually lower correlation, mirroring Burgers’ pattern. This is the first physical evidence that the redundancy mechanism itself, not just the frequency ceiling theory, generalizes across PDEs.

6 Mechanism: Representational Redundancy and Spectral Collapse

This section connects the abstract expressivity ceiling covered in section 5 to an observable, mechanistic account of what happens inside the trained network.

6.1 Post quantum layer redundancy

We measured the mean pairwise correlation between the quantum layer’s measured qubit outputs across an evaluation batch, for all 12 sweep configurations:

configuration	$r=1$	$r=2$	$r=3$	$r=5$
$q3$	0.991	0.906	0.428	0.823
$q4$	0.996	0.668	0.869	0.617
$q5$	0.983	0.919	0.881	0.889

Table 2: Post-quantum-layer mean pairwise correlation

The $r=1$ step is clean: every configuration sits at near-total redundancy (0.983-0.996), with little qubit-count dependence. *The claim that correlation decreases smoothly with reupload count does not hold at full resolution:* the $r=2$, $r=3$, and $r=5$ configurations span a wide band (0.43-0.92) with no clean order. Notably, $q3, r=3$ (0.428) is **lower** than $q3, r=5$ (0.823), the inverse of what the four-corner comparison alone would suggest. This is not a computation artifact: $q3, r=3$ also has the second-highest effective rank among $q3$ configurations (Section 6.1.1), so two independent metrics agree it is a real, low-redundancy configuration.

A corrected, defensible claim is a step function at $r=1$ followed by a noisy plateau at $r \geq 2$, not a smooth decrease across all tested reupload counts. The qualitative core of the mechanism remains solid, however: pre-quantum-layer (classical encoding) activations are near-zero correlated for *all* 12 configurations, ruling out the possibility that any of this redundancy is inherited from the input rather than produced by the quantum layer.

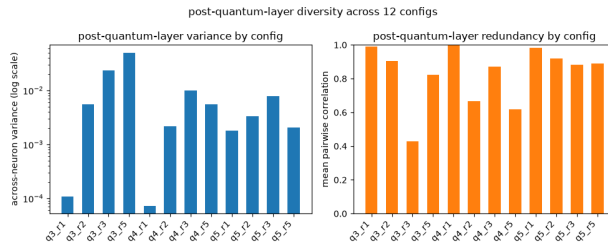


Figure 2: Post quantum layer variance (on a log scale) and pairwise correlation between all 12 sweep configurations. $q3, r=3$ ’s outlier low correlation is visible directly, as is the $r=1$ step and the noisy $r \geq 2$ plateau.

6.1.1 Independent confirmation via effective rank.

As a second, independently sourced diagnostic, we computed the effective rank (via singular value decomposition of the post-quantum-layer output, using the entropy-based definition of O. Roy and M. Vetterli

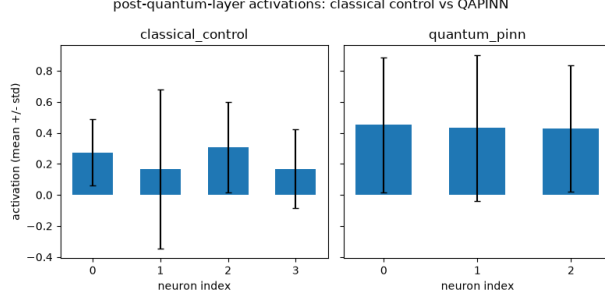


Figure 3: Post quantum layer neuron activations, classical control vs QAPINN, mean $\pm$ std per neuron. Shows original uniformity of the 4 neurons observed that motivated the redundancy investigation.

2007)[8] for across all 12 sweep configurations:

Configuration	$r=1$	$r=2$	$r=3$	$r=5$
$q3$	1.37	2.21	2.89	2.50
$q4$	1.26	2.31	2.25	3.00
$q5$	1.69	2.16	3.21	2.82

Table 3: effective rank across 12 sweep configurations

This confirms, at full resolution, the same pattern as Section 6.1: a clean rise from $r=1$ (1.26-1.69, near the theoretical minimum) to $r=2$ (2.16-2.31), followed by a plateau at $r=3$ and $r=5$ rather than continued growth. Average effective rank is $r=1$: 1.44, $r=2$: 2.23, $r=3$: 2.78, $r=5$: 2.77; essentially flat past $r=2$. The single highest value in the dataset is $q5, r=3$ (3.21), not $q5, r=5$, mirroring the same non-monotonicity seen in Section 6.1. We note this as an *application* of a recently published metric, not one of our own; the two independent observations agreeing on where the flattening starts gives us confidence it reflects a real property of the trained circuits rather than noise in either measurement.

6.2 Cross-PDE redundancy check

The same diagnostics computed on the heat equation (Section 5.5) reproduce the redundancy signature across both seeds, with one notable instability: $q3, r=5$ swings from 0.921 (seed 0) to 0.630 (seed 1), the largest seed-to-seed swing in the heat-equation redundancy data, echoing the seed-instability pattern in Section 8.4.

6.3 Measurement-artifact check

Since every qubit was measured with the same observable (Pauli-Z), we checked whether the observed uniformity was a measurement artifact by re-measuring with mixed Pauli observables (Z, X, Y). The result depends on configuration: for the low-expressivity configuration ($q3, r=1$), the apparent uniformity is almost entirely artifact (activation spread increases $31\times$ under mixed measurement). For the high-expressivity configuration ($q5, r=5$), it is only partially artifact (spread increases $1.9\times$), indicating genuine observable-independent structure. We report this as a mixed result rather than forcing a single conclusion; the *correlation-based* redundancy metric of Section 6.1, computed under identical measurement for all configurations, remains the reliable finding.

6.4 The mechanism, in order

Low reupload count $\rightarrow$ low accessible Fourier ceiling (Section 5.1) $\rightarrow$ the quantum layer can express only around one frequency component $\rightarrow$ the measured qubits, sharing that limited expressivity, produce near-identical output values (Section 6.1) $\rightarrow$ the classical post-layer receives practically a rank-1 signal $\rightarrow$ the

network cannot represent the higher-frequency structure the true solution needs. This is a specific, testable mechanism linking an architectural property to an observable interpretability signature to an accuracy consequence, and we regard it as our strongest contribution.

6.4.1 Layer level attribution via Shapley values

As a further diagnostic, we computed the exact Shapley values [9] for the five reupload layers of the $q5, r=5$, we treat each as a "player" and the relative L_2 error improvement from a subset of layers as the value function. We use full enumeration over all 32 subsets rather than Monte Carlo sampling.

We verified the correctness before interpreting the result: the full-circuit ($S = \{0, \dots, 4\}$) ablation reproduces the checkpoint's already recorded L_2 (0.594387) as an exact match, confirming the ablation mechanism is implemented correctly. The five shapley value sum to exactly 1.011517, which exactly matches the full circuit vs empty circuit L_2 gap ($1.605904 - 0.594387$), the expected efficiency property of exact Shapley values, confirming no arithmetic error. Average L_2 error improves smoothly as more layers are added to a subset (singles ≈ 1.0 – 1.1 , pairs ≈ 0.86 – 0.99 , triples ≈ 0.75 – 0.83 , quadruples ≈ 0.74 – 0.82 , full circuit 0.594), without any erratic jumps, which is consistent with layers contributing incrementally.

Layer	Shapley value
0	0.182
1	0.168
2	0.235
3	0.208
4	0.219

Table 4: Shapley value

Mean ≈ 0.202 , range 0.168-0.235 ($\approx 1.4\times$ spread, $\approx 11\%$ relative standard deviation): *broadly uniform with mild variation*, not exact equality (the maximum is 16% above the mean, the minimum 17% below, with no clean trend by layer position). This is consistent with Section 5.1.1's framework, where each reupload layer contributes one interchangeable unit of Fourier frequency content; roughly uniform attribution is exactly what we would expect, modulo sample size (five layers, one checkpoint, no repeats). One detail we flag without over-interpreting: layer 2 (the middle layer) attributes highest and layer 1 lowest, which could reflect a positional effect from the CNOT ring's chain structure or could simply be noise at $n = 5$.

6.5 Entanglement pattern variation

Holding $q5, r=5$ fixed and varying only the entangling topology (ring, the default; a linear CNOT chain; no entanglement) produces a result that runs counter to the direction that Section 6's redundancy accuracy relationship would predict:

Topology	L_2	PDE residual	Correlation	Effective rank
Ring (default)	0.625	0.0359	0.889	2.82
Linear	0.640	0.0284	0.024	2.89
None	0.743	0.0194	-0.070	4.52

Table 5: Entanglement pattern variation, $q5, r=5$.

Removing entanglement collapses the redundancy signature almost entirely, exactly what Section 6's mechanism would predict should correlate with better accuracy. Instead, L_2 gets worse as entanglement is removed: ring (highest redundancy) has the best fit, and none (lowest redundancy, near max effective rank) has the worst. This is the opposite direction from the reupload count driven redundancy in Sections 6.1-6.3.

We see this as evidence that low correlation between measured qubits is not a single uniform phenomenon. When redundancy decreases because reupload count increases, qubits access more Fourier content, a productive decorrelation. When redundancy decreases because entanglement is removed, qubits compute more

independently, without the joint structure the classical post layer’s training relies on, an unproductive decorrelation. The correlation metric can not distinguish these alone; the accuracy outcome can though. This is a limitation of representational redundancy as a standalone diagnostic. the PDE-residual/ L_2 anti-correlation (Section 6.9) does reproduce here (residual improves from 0.036 to 0.019 as L_2 worsens), a third independent confirmation of that specific pattern.

6.6 Measurement operator variation

Replacing per qubit Pauli-Z expectation values with the full measurement probability vector (`qml.probs()`, output dimension $2^{n_{\text{qubits}}}=32$ rather than $n_{\text{qubits}}=5$) on $q5, r=5$ raises the post layer parameter count from 71 to 98, breaking the parameter matched comparison made in Section 9. L_2 worsens ($0.625 \rightarrow 0.751$) while PDE residual improves ($0.0359 \rightarrow 0.0068$), this is a third confirmation of the decorrelation pattern under an independent circuit modification. Redundancy metrics on the probs output (correlation 0.168, effective rank 4.24 on a 32-dimensional output) are not directly shape comparable to the 5-dimensional Pauli-Z configurations.

6.7 Loss landscape sharpness at the converged minimum

The fact that this checkpoint sits at a genuine local minimum (loss 0.115 at zero perturbation along all three directions, rising monotonically in both directions along each axis) is confirmed by a perturbation slice along three random directions in parameter space [10], centered on the trained $q5, r=5$ checkpoint. The sharpness varies by direction: loss rises to 0.31-0.42 at the largest tested perturbation, roughly a 3-4 $\times$ increase, direction 0 sharpest, direction 2 shallowest.

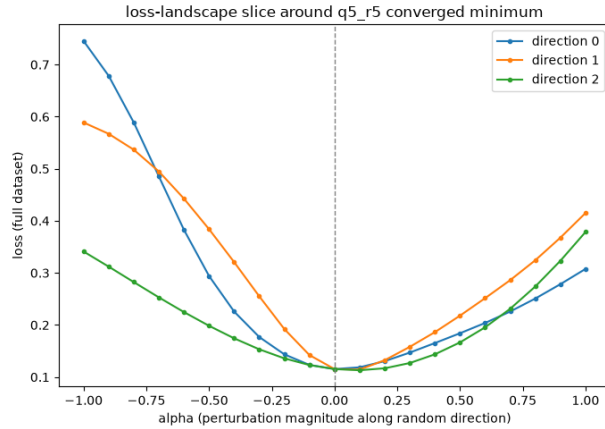


Figure 4: Loss-landscape slice around the converged $q5, r=5$ minimum, three random directions.

6.8 Redundancy evolves during training

While tracking correlation and effective rank at epochs 100, 300, and 500 for $q3, r=1$ and $q5, r=5$ it was shown that redundancy is not fixed at initialization, instead it erodes steadily during both configurations.

Config, metric	Epoch 100	Epoch 300	Epoch 500
$q3, r=1$, correlation	0.991	0.884	0.783
$q3, r=1$, effective rank	1.37	2.10	2.33
$q5, r=5$, correlation	0.889	0.799	0.772
$q5, r=5$, effective rank	2.82	3.30	3.50

Table 6: Redundancy tracked over training epochs.

Both configurations move in the same direction, decreasing correlation while increasing effective rank, as training proceeds. The network learns to use more of its available representational capacity over time; the mechanism listed in Section 6.4 describes the asymptotic structural ceiling, but approach to that ceiling is itself a training dynamics phenomenon (still measured on only two configurations)

6.9 Confirmation via prediction sufficiency (PDE residual vs. L_2 error)

Across all 12 sweep configurations, mean PDE residual error and mean L_2 error are **anti-correlated** (Pearson $r = -0.970$ on the 12 configuration-level means). The worst-fitting configuration ($q3, r=1$: $L_2 = 0.822$) has the *lowest* residual error; the best fitting configuration ($q5, r=5$: $L_2 = 0.625$) has the *highest* residual error, roughly $15\times$ higher. This holds at the individual-run level too: across all 36 raw seed-level runs, $r = -0.923$, confirming the effect is not an artifact of averaging away seed noise.

This is the signature of configurations leaning toward an over-smoothed solution, with a specific explanation for a shock-forming, convection-dominated PDE like Burgers’: since the residual is $u_t + uu_x - \nu u_{xx}$, a smooth, low derivative function trivially satisfies it almost everywhere. A higher capacity network capable of representing the true shock has large u_x and u_{xx} where the true solution does, so any residual mismatch there is amplified by the derivative terms in the PDE residual itself, which is why better L_2 fit and better residual satisfaction pull in opposite directions here. This gives a quantum layer specific mechanistic cause for the general spectral bias failure mode already documented in the classical PINN literature.

We also checked this on the extended-training dataset (Section 8), spanning epochs 200-500 across all four corner configurations (both seeds) and the parameter-matched classical control: the anti-correlation remains strong ($r = -0.905$, $n = 36$, the complete dataset), close to the original sweep’s $r = -0.970$. We further confirmed this by measuring solution flatness directly:

	Total variation	Mean gradient	Amplitude range
$q3, r=1$ (worst L_2 , lowest residual)	0.677	0.339	0.598
$q5, r=5$ (best L_2 , highest residual)	1.560	0.780	0.941
Cole-Hopf reference	3.688	1.844	2.000

Table 7: Solution flatness, seed-0 checkpoints.

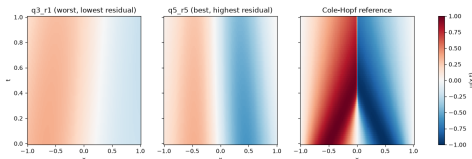


Figure 5: $u(x,t)$ heatmaps for $q3, r=1$ (worst, lowest residual), $q5, r=5$ (best, highest residual), and the Cole-Hopf reference. The reference’s sharp shock front is visually absent from both trained configurations, more so for $q3, r=1$.

7 Trainability: No Barren-Plateau Signal at this Level

We measured the gradient variance of the PDE-residual loss with respect to the quantum layer’s variational parameters, we did this across 50 random initializations per configuration. Across $n_{\text{qubits}} \in \{3, 4, 5\}$, values remain in the 10^{-6} - 10^{-5} range with no systematic collapsing trend as qubit count increases, this result was expected with the absence of barren plateaus. However, one of the configurations ($q3, r=5$) is a confirmed outlier (mean gradient variance $\approx 1.7 \times 10^{-4}$, 15 - $40\times$ higher than all others); we verified and reported this as a reproducible outlier via independent reruns.

We conclude that at the qubit scales we tested (3-5 qubits), **expressivity, not trainability, is the operative constraint.**, as barren plateaus usually form at much larger qubit counts, while expected, it

functions as a necessary control for the disentanglement claim in Section 5, rather than just as a standalone headline result.

8 Convergence Dynamics: Expressivity vs Training budget

8.1 Original (100-epoch) sweep result.

Averaged across the sweep, L_2 error by reupload count: $r=1$: 0.867; $r=2$: 0.729; $r=3$: 0.690; $r=5$: 0.671, this shows a clear trend of diminishing returns. By qubit count: $q3$: 0.773; $q4$: 0.721; $q5$: 0.723, a minor gain from three to four qubits, then a saturation from four to five. At face value, this seemed to confirm a straightforward expressivity ceiling situation.

8.2 Extended training reveals the gap is mostly a convergence-speed effect.

We extended training to 500 epochs (from 100) to see the sweeps extreme corners, under plain gradient descent, and ran a second seed on all four corners once each primary result landed. $q3, r=5$'s second seed was the last to land; once we got it, the seed robustness picture (Section 8.4) is different from what the three corner subset suggested. Results:

Configuration	L_2 @ 100 epochs	L_2 @ 500 epochs, seed 0	L_2 @ 500 epochs, seed 1
$q3, r=1$ (worst)	0.822	0.638	0.640
$q5, r=1$	0.759	0.632	0.640
$q3, r=5$	0.694	0.638	0.585
$q5, r=5$ (best)	0.625	0.594	0.619

Table 8: Extended training

At 100 epochs, the gap between the best and worst tested configuration is ≈ 0.28 . At 500 epochs, the two $r=1$ corners converge tightly to a 0.632-0.640 band, consistent with the convergence speed story. The two $r=5$ corners ($q3, r=5$, $q5, r=5$) do not join that band: across their four seed configuration combinations they span a much wider 0.585-0.638 range, with $q3, r=5$'s second seed (0.585) landing *below* anything either $r=5$ corner reached previously. The gap between best and worst shrinks substantially (from ~ 0.28 to ~ 0.05 at the band level) once training budget is no longer a confound; the headline convergence-speed effect holds, but the residual ceiling effect tracks re-upload count on its own, not the qubits-and-reuploads interaction a partial dataset once suggested; see Section 8.4.

8.3 Revised Claim

Reupload count primarily controls the **optimization convergence speed**; a residual, more variable **achievable accuracy ceiling** effect is associated with a high reupload count, not with qubit count. There is a caution which must be held whenever training, being that a fixed shallow training budget substantially overstates the role of circuit expressivity in QML sweeps.

8.4 Seed robustness

With $q3, r=5$'s second seed and $q5, r=5$'s third seed both landed, the full seed-spread table is:

This revises a previously made claim in the report An initial reading, based on three of four of the corners, attributed instability to $q5, r=5$'s combination of maximal qubit count *and* maximal re-upload count. However, the completed four corner table does not support that, as $q3, r=5$, which maximizes *only* re-upload count, shows a *larger* spread (0.053) than $q5, r=5$ (0.044). A corrected, simpler claim is that seed instability tracks **reupload count alone**: both $r=5$ corners are unstable, both $r=1$ corners are stable, independent of qubit count. This is more consistent with the qubit independence finding of Section 5.1. The qualitative finding that higher expressivity configurations are less seed reproducible, is better supported; the proposed mechanism for it, an interaction requiring both dimensions, does not survive and is removed.

Corner	Seeds	Spread
$q3, r=1$	0.638, 0.640	0.003
$q5, r=1$	0.632, 0.640	0.008
$q3, r=5$	0.638, 0.585	0.053
$q5, r=5$	0.594, 0.619, 0.638	0.044

Table 9: Seed spread, all four corners, complete data.

9 Classical vs. Quantum Comparison

Our original classical control (17 trainable parameters, plain gradient descent, 100-epochs) reached $L_2 = 0.895 \pm 0.005$; every tested QAPINN configuration outperformed it, however that comparison was confounded on two axes, being parameter count and optimizer choice, which we correct here with a properly matched final comparison.

9.1 The fair-comparison table

Model	Optimizer	Epochs	Batch	Params	L_2	Converged?
Classical (matched)	Adam	5000	200	73	0.635	Yes, flat from ep. 1500
Classical (deep/wide)	Adam	5000	20	3021	0.499	Descending, noisily
Classical (deep/wide)	Adam	5000	200	3021	0.436	Yes, clean plateau
Quantum $q5, r=5$	GD	500	20	71	0.594-0.638	Yes, oscillating plateau
Quantum $q5, r=5$	Adam	1000	20	71	0.608	Yes, oscillating plateau

Table 10: Fair classical-vs-quantum comparison.

9.2 The batch size caveat

Quantum could not feasibly run at batch=200 within the submission timeline (~ 120 s/epoch at that batch size, ~ 7 days for a full run); all quantum results use batch=20. The batch=200 classical numbers (0.436, 0.635) are **not on equal footing** with quantum. The only comparison in this table that is matched with the optimizer, epoch budget, *and* batch size is classical (deep/wide, batch=20) at $L_2 = 0.499$ against quantum’s 0.594-0.638.

9.3 Headline finding

Under the one comparison, classical outperforms every tested quantum configuration: 0.499 vs. 0.594-0.638, roughly 16-20% relative improvement for classical. The apparent quantum advantage documented earlier in this report was an artifact of comparing a reasonably configured quantum layer against a badly under trained classical baseline.

9.4 Optimizer asymmetry

For classical, optimizer and capacity both matter: matched-capacity classical under Adam reaches 0.635, a deeper/wider network under Adam reaches 0.436-0.499. For quantum, neither optimizer choice nor tested capacity moved the needle, this happens across two optimizers and both seeds, every $q5, r=5$ run lands in a 0.594-0.638 band. Quantum’s Adam boost was nowhere near classical’s, consistent with an accuracy ceiling set by the fixed Fourier structure (Section 5) and the redundancy it produces (Section 6), which optimizer choice would not be expected to change much.

9.5 Wall-clock cost

Quantum $q5, r=5$ under GD took 12324.4s (3h 25m) to complete 500 epochs; under Adam, 9794.8s (2h 43m) to reach epoch 300, where L_2 first entered its plateau band. Classical (matched, Adam) reaches its epoch-1500 plateau, directly measured, at 1786.1s (≈ 29.8 minutes). Quantum’s wall-clock penalty is roughly **5-7 $\times$** classical’s time-to-plateau, not an order of magnitude. We flag this as it is not a completely one to one timing comparison either: classical’s figure is at batch=200 (10 $\times$ more collocation points per epoch than quantum’s batch=20), so classical does more work per epoch and still reaches its plateau faster in absolute terms, which strengthens our finding.

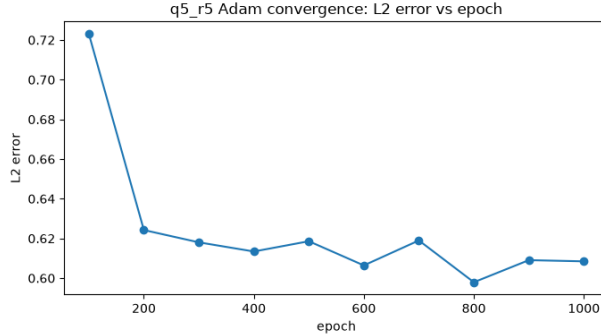


Figure 6: $q5, r=5$ under Adam, L_2 vs. epoch, full trajectory.

9.6 What this means for the project as a whole

The mechanistic findings of Section 6 do not depend on which architecture wins the comparison, and are strengthened by it: they explain more than just how the quantum layer’s expressivity ceiling manifests, but why it persists across optimizers and seeds in a way classical’s performance does not.

10 Design Methodology

For a target PDE and a specific QAPINN architecture, we propose the following procedure, derived directly from Sections 5-9:

1. Compute the Fourier spectrum of the target solution across the domain, identifying required bandwidth per region.
2. Compute the circuit’s accessible frequency ceiling from its encoding scheme (Section 5.1), converting to physical units via the pre-layer’s local derivative from a fixed initialization (Section 5.2).
3. Compare required bandwidth to accessible ceiling per region using relative-distance-from-sufficiency, not a hard threshold, to predict where additional re-uploads yield the largest relative accuracy gains (Section 5.4).
4. Check gradient variance across candidate qubit counts to confirm trainability is not the binding constraint (Section 7).
5. Before attributing an accuracy gap to expressivity, verify it survives a substantially longer training budget under a well-tuned optimizer (Section 8); a shallow, fixed-budget comparison risks conflating convergence speed with expressivity ceiling.
6. Use post-quantum-layer representational redundancy as a diagnostic, but do not assume its direction of effect transfers to every circuit modification: redundancy driven by re-upload count and redundancy driven by entanglement topology have opposite accuracy consequences (Section 6.5).

7. Before any classical-vs-quantum comparison, match optimizer, epoch budget, and batch size; an unmatched comparison will overstate the quantum architecture’s advantage (Section 9).

11 Limitations and Recommended Next Steps

- Both seeds landed for the heat equation, but a few configurations ($q5, r = 1$ and $q5, r = 5$) still show seed to seed movement rather than a convergence, a third seed should settle this.
- Extended-training results have two seeds for all four corners; the four-corner ranking and the exact classical-vs-quantum gap are not established beyond this seed count, only the qualitative direction of the findings should be treated as settled.
- Representational correlation does not have a uniform relationship to accuracy across all circuit modifications (Section 6.5): productive when redundancy is driven by re-upload count, unproductive when driven by entanglement topology. This metric should not be assumed to transfer without checking accuracy directly on any new circuit variant.
- No configuration reaches expressivity sufficiency in either the smooth or shock region; read the sufficiency numbers as relative distance, not pass/fail.
- The qubit-count range tested (3-5) is small; the ”no barren plateau” finding is expected, not surprising, at this scale.
- Simulator only; nothing here has touched real quantum hardware.

11.1 Recommended next steps

1. Investigate why redundancy has two different accuracy relationships depending on cause (Section 6.5), the least-understood finding in the project; a systematic sweep across entanglement topologies beyond one configuration would clarify whether this is general or specific to $q5, r=5$.
2. Extend Shapley-value attribution beyond a single configuration and checkpoint, and to gate-level rather than layer-level.
3. Extend the qubit-count range beyond 3-5 to determine where barren-plateau effects begin dominating.
4. Add a third seed to all four extended-training corners to fix group means and spreads with full confidence.

12 Use of Generative AI

All research questions, hypotheses, theoretical framing, and interpretive judgments in this work are the authors’ own. Generative AI (Claude Code) was used for pipeline execution and implementation support: writing and running training/evaluation scripts, executing the configuration sweep, and running diagnostic analyses. It was not used to originate hypotheses, decide what the data means, or determine which findings to report. All AI-assisted output was reviewed and verified against source data before being trusted. The authors remain fully responsible for the accuracy, originality, and quality of all submitted work.

13 Team Contributions

This Project was carried out under **Qinetic Research Lab**

- **Jithesh Mithra: Head Researcher.** Primary contributor to this project: designed and executed the training/evaluation pipeline, the full configuration sweep, and the majority of the empirical analysis. Led the overall project direction, owner of repository.

- **Isaac Leon: Theory Contributor.** Contributed to the theoretical framing, including work on the Fourier-ceiling analysis and the expressivity-vs-trainability disentanglement synthesis, and identified the PDE-residual/ L_2 anti-correlation discussed in Section 6.9.

Both contributors reviewed the overlay analyses connecting theoretical predictions to empirical results (Sections 5-8) to guard against one-sided interpretation of ambiguous findings.

14 Conclusion

We set out to explain when, why, and how a quantum layer changes the learning dynamics of a PINN solving Burgers’ equation, not to show that it outperforms a classical baseline. The accessible Fourier frequency spectrum of the quantum layer is set by the data-reupload count, independent of qubit count, derived directly from the encoding gate structure. This ceiling manifests as measurable representational redundancy among the quantum layer’s outputs, mechanistically tied to a spectral-bias-like collapse toward over-smoothed solutions, confirmed by two independent metrics on two PDEs (Burgers’ and the heat equation). Trainability is not the binding constraint at the qubit scales tested.

We also find that most of what a shallow sweep attributes to expressivity differences is a convergence-speed effect that erodes under longer training, and that redundancy’s relationship to accuracy is not uniform: the same low-correlation signature is productive when driven by reupload count and unproductive when driven by removing entanglement.

Once classical and quantum architectures were compared under the same conditions (optimizer, epoch budget, collocation batch size), **classical outperforms quantum on this problem, at this scale, with this architecture** ($L_2 = 0.499$ vs. $0.594\text{--}0.638$), at a fraction of the wall-clock cost. We regard the mechanistic findings the Fourier ceiling derivation, the redundancy signature and its complications, and the spectral-bias connection as this work’s primary contribution, since they hold regardless of which architecture is more accurate. The self-corrected documentation across Sections 8 and 9, which initially favored quantum before revision, is itself part of the finding: a direct illustration of the fixed-budget, unmatched-baseline overstatement this report’s methodological caution warns against.

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